

git repack is a Git command that packs unpacked objects in a repository into one or more **pack** files. It helps reduce disk space and can reorganize packs into a more efficient layout.  graphapp.ai  git-scm.com

Overview of Git Repack

git repack is a command used in Git to optimize the storage of a repository by consolidating unpacked objects into one or more **pack** files. This process is essential for maintaining the efficiency and performance of Git repositories, especially those with a large number of loose objects.

Purpose of Git Repack

- **Reduce Disk Space:** By packing loose objects into a single file, git repack significantly decreases the amount of disk space used by the repository.
- **Improve Performance:** Consolidating objects can enhance the speed of operations that read from the disk, making the repository more responsive.
- **Organize Data:** It can reorganize existing packs into a more efficient layout, which is beneficial for both local and remote operations.

Key Features

Feature	Description
Pack Unpacked Objects	Combines all unpacked objects into a new pack file.
Optimize Storage	Reduces redundancy by consolidating loose objects and fragmented packs into fewer, optimized packs.
Support for Filters	Allows filtering of objects based on specified criteria during the packing process.
Configuration Options	Offers various flags to control the packing process, such as depth and window size for delta compression.

Usage Scenarios

- **Housekeeping:** Regularly running git repack helps maintain a clean and efficient repository.
- **Repository Transfer:** Preparing a repository for transfer to another system can be done more efficiently by packing all objects into a single file.
- **Performance Optimization:** In high-activity repositories, using git repack can help manage the number of pack files and improve overall performance.

By utilizing git repack, users can ensure their Git repositories remain efficient and manageable, especially as they grow in size and complexity.

Explore More

What are the benefits of using git repack for repository management? ✓

How does git repack compare to other Git commands for optimizing storage? ✓

What are the potential risks of using git repack in a collaborative environment? ✓

Ask a follow-up with Duck.ai

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Git

<https://git-scm.com/docs/git-repack>

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Git - git-repack Documentation

`git repack` ensures this by determining a "cut" of packfiles that need to be repacked into one in order to ensure a geometric progression. It picks the smallest set of packfiles such that as many of the larger packfiles (by count of objects contained in that pack) may be left intact.

Topics

By default, the command passes `--delta-base-offset` option to `git pack-objects`.



Git

<https://git-scm.com/docs/git-repack/2.8.6>

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Git - git-repack Documentation

By default, the command passes `--delta-base-offset` option to `git pack-objects`; this typically results in slightly smaller packs, but the generated packs are incompatible with versions of `Git` older than version 1.4.4. If you need to share your repository with such ancient `Git` versions, either directly or via the `dumb-transport`...



Stack Overflow

<https://stackoverflow.com/questions/28720151/git-gc-aggressive-vs-git-repack>

...

github - git gc --aggressive vs git repack - Stack Overflow

Linus ends his mail with the conclusion that `git repack` with a large `--depth` and `--window` is the better choice in most of time; especially after you imported a large project and want to make sure that `git` finds good deltas.



MIT - Massachusetts Institute of Technology

<https://web.mit.edu/git/git-doc/git-repack.html>

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git-repack (1) - MIT - Massachusetts Institute of Technology

Aug 19, 2026 · `git repack` ensures this by determining a "cut" of packfiles that need to be repacked into one in order to ensure a geometric progression. It picks the smallest set of packfiles such that as many of the larger packfiles (by count of objects contained in that pack) may be left intact.



Atlassian Support

<https://support.atlassian.com/bitbucket-data-center/kb/understanding-git-repack-and-git-prune/>

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Understanding git repack and git prune options in Bitbucket Data...

Jan 30, 2026 · Discover how Bitbucket Data Center manages repository maintenance using `git repack` and `git prune` commands.



Linux Commands

<https://commandlinux.com/man-page/man1/git-repack>

...

GIT-REPACK - commandlinux.com

Jul 27, 2026 · If you need to share your repository with such ancient `Git` versions, either directly or via the `dumb http` or `rsync` protocol, then you need to set the configuration variable `repack.UseDeltaBaseOffset` to "false" and `repack`.



Mastering Git Repack: Optimize Your Repository Easily

`git repack` is a **Git** command that is designed to optimize the storage of data in a **Git** repository by packing multiple loose objects into a single pack file. This process helps to decrease the overall size of the repository and improve performance when it comes to operations like cloning, fetching, and pulling.



Git Cookbook - git-repack (1)

By default, the command passes `--delta-base-offset` option to `git pack-objects`; this typically results in slightly smaller packs, but the generated packs are incompatible with versions of **Git** older than version 1.4.4. If you need to share your repository with such ancient **Git** versions, either directly or via the dumb...



git-repack (1) - Linux manual page - man7.org

If the tip layer contains fewer packs than `repack.midxNewLayerThreshold`, those packs are excluded from the geometry entirely, and a new layer is created for any new pack (s) without disturbing the existing chain. `--name-hash-version=<n>` Provide this argument to the underlying `git pack-objects`...

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